Market Access and the Production of Information

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ABSTRACT

Partial access can induce foreign institutional investors, efficient processors of data without soft information, to allocate scarce processing capacity to shares they cannot hold, because the acquired information affects prices of shares they can trade. Soft information, data-processing technology, market access and portfolio restrictions interact through equilibrium prices to determine information production. Market access can raise price informativeness while lowering the average posterior precision. A China calibration explains why a small foreign market share can play a disproportionate informational role. Expanding market access in China has little impact, while relaxing the short-sale constraint substantially improves price informativeness.

Keywords: market access; information acquisition; price informativeness; cross-asset learning; financial liberalization

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Opening a domestic stock market to foreign investors changes who can hold domestic risks and who finds it valuable to produce information about that risk. This paper studies that information-production channel of market access: when investors differ in their information endowments and data-processing technologies, access restrictions shape the production of information.

Domestic investors alone possess soft information generated by local knowledge and relationships, while hard information can be acquired by both foreign and domestic institutions by processing data, where foreign institutions have better data-processing technology. Thus neither group is uniformly better informed: domestic investors benefit from soft information, whereas foreign institutions have a processing advantage. This heterogeneity is embedded in a noisy rational-expectations model with endogenous information acquisition, where institutions allocate scare data-processing capacity across assets before trading.

The model shows how market access changes information production when investors differ jointly in processing technology, soft information, and market access. It delivers three central theoretical implications. Firstly, market access and information relevance need not coincide: investors can rationally learn about assets they cannot hold. Secondly, cross-asset learning occurs only at intermediate precisions: information that is too noisy is not worthy to learn, while information that is sufficiently precise is already reflected in equilibrium prices. Thirdly, market access can raise price informativeness while reducing average investor information; making an asset investable can also reduce its price informativeness.

private information acquisition and price informativeness are distinct equilibrium objects. Foreign entry can add hard information while diluting local soft information, and portfolio constraints can prevent private information from being incorporated into prices.

Under partial access, foreign institutions can trade only one asset type. Because fundamentals of assets are correlated, however, information about an inaccessible asset can improve valuation of an accessible one. Market access therefore determines not only accessible assets but also the return to information production.

The mechanism is inherently general equilibrium. Foreign information changes informed demand and hence the price signal observed by domestic investors. Domestic institutions then reallocate their own processing capacity in response to that signal. The resulting feedback determines both what investors learn privately and how much of that information is embedded in prices. This distinction matters because a policy that increases the information held by some

investors need not improve the information revealed to the market as a whole.

The model also clarifies the role of nontradable assets. Under partial access, foreign institutions cannot take positions in Type-II shares, but Type-II information can still have value for Type-I trading. The relevant object is not whether an investor can hold an asset, but whether its payoff contains information useful for valuing an asset the investor can hold. This cross-asset channel gives market access a broader effect than a one-asset model would imply: opening one segment can affect information acquisition and prices in another segment that remains closed.

Portfolio constraints add a separate margin. With a short-sale constraint, information affects whether an investor participates in a market as well as the position taken conditional on participation. Constraints can therefore weaken the incorporation of private information into prices even when investors have strong incentives to acquire that information. This channel is important for interpreting liberalization counterfactuals: expanding the set of assets foreigners may hold and relaxing their trading constraints are economically distinct interventions.

The model is calibrated to the Shanghai–Hong Kong Stock Connect program. The calibration addresses why a small foreign ownership share can have a disproportionate role. Ownership is not a sufficient statistic for informational influence: even a small foreign sector can affect the public price signal and thereby the information choices and portfolios of the larger domestic sector. The benchmark is conditional on a maintained processing-technology advantage; the associated identification and sensitivity analysis is reported in the Internet Appendix.

The full-access counterfactual separates investability from trading flexibility. At the estimated Chinese benchmark, expanding foreign investability from partial to full access changes little when short-sale constraints remain in place. Relaxing the short-sale constraint, however, substantially improves price aggregation. Thus portfolio flexibility, rather than incremental investability alone, is the quantitatively dominant policy margin in this calibration.

The quantitative analysis is intended to discipline the model's equilibrium mechanism, not to infer that foreign ownership mechanically causes large market-wide effects. In the benchmark, foreign institutions are small in both investor mass and observed ownership, and their positive positions are infrequent under the short-sale constraint. Their informational role can nevertheless be consequential because information affects marginal price-setting demand and is transmitted through a public price signal to the larger domestic sector. The calibration therefore distinguishes ownership, investor mass, participation, and information production rather than treating any one of these objects as a sufficient statistic for the others.

I complement the structural analysis with reduced-form evidence around the Stock Connect announcement and launch. Eligible shares exhibit differential movements in return-based information outcomes, but the evidence is descriptive rather than a causal test of the model mechanism.

The empirical setting is useful because Stock Connect created a clear difference between shares that became foreign-investable and those that remained inaccessible under partial opening. The reduced-form evidence documents changes in the information environment around this institutional change, whereas the structural model maps differences in investor technology, soft information, and investability into equilibrium information allocation. Together, the two exercises provide complementary evidence rather than substitutes for one another.

More broadly, the analysis cautions against equating liberalization with a monotone improvement in either private information or price informativeness. The effect of access depends on who enters, what information they can produce, which assets they can trade, and whether their information can be translated into risk-bearing demand. These distinctions are particularly relevant in settings where foreign participation is limited, information technologies differ across investors, and policy reforms relax access and trading restrictions sequentially rather than simultaneously.

The paper makes five contributions. Fourth, the calibration explains why a small foreign ownership share can have a large equilibrium effect. Fifth, the full-access counterfactual separates the effects of investability and trading flexibility and yields the paper's policy implication.

The remainder of the paper proceeds as follows. Section I briefly reviews the related literature. Section II sets the general modelling framework, while Section III derives the information-allocation-aggregation mechanism across market-access regimes and Section IV introduces portfolio constraints and endogenous participation. Section V studies welfare. The quantitative section calibrates the model to Stock Connect, evaluates counterfactual policies, and presents reduced-form evidence. The final section concludes. Proofs of the main results are presented in the Appendix, with additional derivations and empirical detail in the Internet Appendix.

I. Literature

A large literature studies market segmentation, risk sharing, valuation, and the cost of capital around liberalization episodes (Henry, 2000, 2003; Bekaert and Harvey, 2000; Kim and Singal, 2000; Chari and Henry, 2004). Related research has documented changes in domestic informa-

tion environments, including analyst activity, disclosure, and the incorporation of firm-specific information into prices (Bae et al., 2006; Chan and Kwok, 2017; Griffin et al., 2010). This paper introduced an information channel to the theory work and makes information production endogenous: access restrictions determine not only who can trade a security but also where heterogeneous investors direct scarce data-processing capacity. This distinction shifts the focus from the post-liberalization level of information to the equilibrium incentives that generate it. Extending market access affect prices not only through risk sharing and capital flows, but also altering the relative return to producing information.

Another line of research has documented empirical evidence on both local-information advantages and foreign advantages associated with sophistication and information processing (Choe et al., 2005; Dvořák, 2005; Hau, 2001; Grinblatt and Keloharju, 2000). This paper theoretically formalizes this multidimensional heterogeneity by separating locally endowed soft information from universally available hard information that can be acquired through processing data (Liberti and Petersen, 2019; Gao and Zeng, 2022; Gao and Huang, 2020). It studies how asset-specific market access governs the interaction of these two advantages. The distinction is important because local information and processing technology need not rank investor types in the same way. Domestic institutions may be better placed to interpret local conditions, while foreign institutions may extract higher precision from a common data source.

The framework builds on noisy rational-expectations models of price aggregation (Grossman and Stiglitz, 1980; Hellwig, 1980; Diamond and Verrecchia, 1981; Verrecchia, 1982) and endogenous information acquisition (Van Nieuwerburgh and Veldkamp, 2009). It introduces asset-specific access restrictions as a source of potential information specialization. Information about an inaccessible asset can benefit investors through its effect on beliefs about accessible asset, creating a wedge between what investors can trade, what they choose to learn, and what prices reveal. The paper is closest to work in which information acquisition and portfolio choice are jointly determined. Its novelty is to make market access an input into that joint problem. The resulting cross-asset learning mechanism differs from conventional diversification motives: investors acquire information about an inaccessible asset not to hold it directly, but because it improves the valuation and trading of an accessible asset.

II. General Framework

To formalize the mechanism, this section lays out the common economic environment used in every theoretical regime.

A. Assets

There is a domestic safe asset with the rate of return r. The equities of domestic firms are originally only traded in the domestic markets. They are modeled as two aggregate risky securities: a type-I security that becomes eligible for international trade once the market opens to foreign investors, and a type-II risky asset that remains closed to international investors under partial opening and becomes investable only under full opening. Period-t dividend $\widetilde{\boldsymbol{d}}$ follows an AR(1) process, $\widetilde{\boldsymbol{d}} = \boldsymbol{\mu} + \boldsymbol{G}\boldsymbol{d} + \widetilde{\boldsymbol{y}}$, where the dividend growth rate $\boldsymbol{G} \equiv diag((g_1, g_2)')$ and the dividend shock $\widetilde{\boldsymbol{y}} \overset{i.i.d.}{\sim} N(0, \boldsymbol{\Sigma}_y)$. We maintain $r > \max\{g_1, g_2\}$, so discounted dividend values are finite and the asset-specific mappings from effective information into absolute signal-to-noise ratios are positive. Write $\boldsymbol{\Omega}_y \equiv \boldsymbol{\Sigma}_y^{-1}$ and let $\mathcal{C}_y \equiv (\boldsymbol{\Omega}_y)_{12}$ denote its off-diagonal cross-precision. In the two-asset Gaussian case, positive fundamental covariance corresponds to $\mathcal{C}_y < 0$. The dividend is paid out at the end of each period. Denote the ex-dividend aggregate firm value by $\widetilde{\boldsymbol{v}}$, then $\widetilde{\boldsymbol{v}} = \boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\widetilde{\boldsymbol{d}}_t$, where $\boldsymbol{\mu}_v = \frac{r}{r-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}$.

B. Investors

There is a continuum of investors. The potential types are λ_I domestic institutional investors, λ_J domestic individual investors, λ_K foreign institutional investors, and λ_H foreign individual investors. Which types are present is regime-specific: let $\mathcal{L}_t^R \subseteq \mathcal{L} \equiv \{I, J, K, H\}$ denote the types active in period t under regime R; the regime-specific active sets used below are stated at the end of this section. For simplicity, the following sections use $l, l \in \{i, j, k, h\}$ to refer to an investor of any type, and $L, L \in \{I, J, K, H\}$ to refer to any type of investors. Investors L are endowed with initial wealth w_L. Which types are active is determined by the access regime alone: the model is a single trading-date economy, and there is no entry or exit across dates. Individual investors form their portfolios only based on security prices, while institutional investors acquire and process data, obtaining private signals about the dividend innovation. All domestic investors are additionally endowed with soft information about the dividend innovation.

All investors have CARA utility over wealth at the end of the trading date, with a risk-aversion coefficient ρ, that is $U_l = -\exp(-\rho \widetilde{w}_l')$, $l \in \{i, j, k\}$[1]. Investors make portfolio choices $\boldsymbol{q}_l$ to maximize their expected utility conditional on their respective information set and subject to a one-period budget constraint

$$w_l' = rw_l + \boldsymbol{q}_l'(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r) \tag{1}$$

The initial wealth w_K of foreign investors is entirely invested in domestic securities markets; their wealth invested in foreign economies is not modeled. The dividend is paid out at the end of the trading date.

C. Market access and feasible portfolios

The economy is a single trading-date model: the model compares the equilibria under different access regimes $R \in \{\text{cl}, \text{pa}, \text{fu}\}$ (closed, partial, and full access) and portfolio feasibility c (with and without a short-sale cosntraint) at the same trading date rather than the outcome of a sequence of periods. Partial access and without a short-sale constraint are unmarked.

Domestic investors I and J can trade both type of risky assets in every access regime, with feasible set $\mathbb{R}^2_+$ and $\mathbb{R}^2$ with and without a short-sale constraint, respectively. Foreign investors are absent under closed access. Under partial access, foreign institutions K can trade only the Type-I asset, so their feasible set is $\mathbb{R} \times \{0\}$ or $\mathbb{R}_+ \times \{0\}$, while foreign individuals are still absent. Under full access, foreign institutions and individuals can trade both assets, with feasible sets $\mathbb{R}^2$ or $\mathbb{R}^2_+$.

D. Information structure

In each period before making the portfolio choices, institutional investors can acquire data to generate private signals about current dividend innovation $\widetilde{\boldsymbol{y}}$. In addition, domestic investors are endowed with soft information[2] about $\widetilde{\boldsymbol{y}}$. The equilibrium price aggregates all private information.

[1] The superscript $+$ refers to the end of the trading date, when the dividend is paid out.

[2] Intuitively, domestic institutional investors should have more soft information than domestic individual investors. However, since it is not a focus of this paper and does not change the structure of the model nor the results, a simplifying assumption that domestic investors all have the same amount of soft information is made.

At different stages of decision, investors have different information sets. When making data-acquisition decisions, domestic institutional investor i has only the prior and the soft information $\boldsymbol{\xi}_i$, denoted $\mathcal{I}_{i^-} = \{\boldsymbol{\xi}_i\}$, while foreign institutional investor k, when active, has no relevant information beyond the prior. Let $\boldsymbol{s}_l$, $l \in \{i, k\}$, denote the private signals generated from data, and $\boldsymbol{s}_p$ denote the price signal. When investor i selects its portfolio, the choice is based on the equilibrium price, the soft information, and the private signal generated from the acquired data, so $\mathcal{I}_i = \{\boldsymbol{s}_p, \boldsymbol{s}_i, \boldsymbol{\xi}_i\}$. Domestic individual investor j does not acquire or process data and makes its portfolio choice based only on the endowed soft information and the equilibrium price, so $\mathcal{I}_j = \{\boldsymbol{s}_p, \boldsymbol{\xi}_j\}$. Foreign individual investors H, when active, hold no soft information and make their portfolio choice based only on the equilibrium price, so $\mathcal{I}_h = \{\boldsymbol{s}_p\}$.

The general equilibrium definition below does not restrict the price mapping. In the unrestricted economies of Section III, the equilibrium price is a linear function of the signals, and since signals are normally distributed, the price can be expressed as a linear function of shocks:

$$\boldsymbol{p} = \boldsymbol{A} + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{C}\widetilde{\boldsymbol{y}} + \boldsymbol{D}\widetilde{\boldsymbol{x}}. \tag{2}$$

Define the signal-to-noise ratio (SNR) to be $\boldsymbol{\zeta} \equiv \boldsymbol{D}_t^{-1}\boldsymbol{C}$. A transformation of the equilibrium price $\boldsymbol{p}$ can generate an unbiased signal about the current dividend shock. In Period t, investors extract a signal $\boldsymbol{s}_p = \widetilde{\boldsymbol{y}} + \boldsymbol{\zeta}^{-1}\widetilde{\boldsymbol{x}}$, and the precision of the price signal is $\boldsymbol{\Omega}_p = \boldsymbol{\zeta}'\boldsymbol{\Omega}_x\boldsymbol{\zeta}$. Under the short-sale regimes of Section IV, exact constrained aggregation makes the global price mapping nonlinear, and that section characterizes its local linear response around a reference state using the same price-signal representation.

Throughout the paper, we distinguish data from information. Data are physically stored on paper or in an electronic device and can be purchased, exclusively or nonexclusively, from data sources. Information is a signal about an unknown random variable, in this case the dividend innovation $\widetilde{\boldsymbol{y}}$, resulting from processing the data. Data can come in different forms, including accounting numbers in a company's annual report (net income, total assets, and long-term debt), daily prices and trading volume, and textual descriptions in a company's news release (e.g., "slightly improves" or "relatively stable"). By processing data, investors obtain a private signal about the dividend innovation. Their data sources, acquired quantities, and data-processing technologies may differ.

Specifically in the model, institutional investors can acquire a total of $\bar{M}$ data points from

two data sources. The data acquisition costs are normalized to zero. Thus investors acquire data at full capacity $\bar{M}$. Denote the numbers of data points acquired by investor l at the trading date for type-I and type-II risky assets as $M_{l,(\mathrm{I})}$ and $M_{l,(\mathrm{II})}$, respectively. Each data source corresponds to one type of firm. The model represents processing technology directly through the quality of the resulting unbiased hard-information signal. A foreign institution that processes a Type-N observation obtains

$$s^{(m)}_{K,(N)} = \widetilde{y}_{(N)} + u^{(m)}_{K,(N)}, \qquad u^{(m)}_{K,(N)} \sim N(0, \Pi^{-1}_{e,(N)}),$$

whereas a domestic institution obtains

$$s^{(m)}_{I,(N)} = \widetilde{y}_{(N)} + u^{(m)}_{I,(N)}, \qquad u^{(m)}_{I,(N)} \sim N\big(0, (\varphi^2 \Pi_{e,(N)})^{-1}\big)\,.$$

Signal noises are independent across observations, investors, and data sources, and $\mathbf{\Pi}_e$ is diagonal. Thus one foreign-processed observation contributes $\Pi_{e,(N)}$ to posterior precision, while one domestically processed observation contributes $\varphi^2 \Pi_{e,(N)}$. For continuity with the technology notation used below, write $\mathbf{\Pi}_e \equiv \beta^2 \mathbf{\Omega}_e$ and $\varphi \equiv \beta_I/\beta \in (0,1)$, where β/β_I indexes the foreign–domestic processing advantage. This is a reduced-form signal-quality wedge: it does not represent domestic institutions as applying an incorrect loading to the same raw observation. The economically operative objects are therefore $(\mathbf{\Pi}_e, \varphi)$.

The resulting information is what we call *hard information*; it can be stored and purchased in the form of *data*, which is then back-engineered into hard information using certain data-processing technology by all domestic and foreign data buyers. However, the other type of information, *soft information*, cannot be stored and distributed as data; it is only held by domestic investors. Assume that domestic investors are endowed with an unbiased signal about the dividend innovation, that is $\widetilde{\boldsymbol{\xi}}_i = \widetilde{\boldsymbol{y}} + \widetilde{\boldsymbol{\epsilon}}_i$, with $\widetilde{\boldsymbol{\epsilon}}_i \overset{i.i.d.}{\sim} N(0, \mathbf{\Sigma}_\epsilon)$. Although domestic institutional investors have an informational advantage with the soft information, they have a technological disadvantage over processing data and are not able to implement the perfect data processing technology. Thus a priori it is unclear who has an overall information advantage.

D.1. Equilibrium

The regime-specific structure can be summarized by two objects: the set of active types $\mathcal{L}^R$ and the feasible portfolio correspondence $\mathcal{Q}_L^R \subseteq \mathbb{R}^2$. At the trading date, $\mathcal{L}^R = \{I, J\}$ if the market access is denied for foreign investors, $\mathcal{L}^R = \{I, J, K\}$ if partial access is granted to foreign institutional investors, and $\mathcal{L}^R = \{I, J, K, H\}$ if full access is granted to all foreign investors. Partial access imposes $\mathcal{Q}_{K,2}^{\text{pa}} = \mathbb{R} \times \{0\}$, whereas full access has $\mathcal{Q}_{K,2}^{\text{fu}} = \mathbb{R}^2$; with a short-sale constraint these become $\mathbb{R}_+ \times \{0\}$ and $\mathbb{R}_+^2$, respectively, and every unrestricted feasible set $\mathbb{R}^2$ is replaced by its nonnegative counterpart $\mathbb{R}_+^2$. For an institutional investor, the feasible allocation of data-processing capacity is $\mathcal{M} \equiv \left\{(M_{(\text{I})}, M_{(\text{II})}) \in \mathbb{R}_+^2 : M_{(\text{I})} + M_{(\text{II})} = \bar{M}\right\}$.

Definition 1 (Market equilibrium). *Fix a market regime R. A rational-expectations equilibrium consists of a price function $\boldsymbol{P}^R(\boldsymbol{d}, \widetilde{\boldsymbol{y}}, \widetilde{\boldsymbol{x}})$, data-acquisition strategies for all institutional investors active in regime R, and portfolio strategies $\{\boldsymbol{q}_l^R\}_{l \in \mathcal{L}^R}$, such that:*

(i) Each institutional investor chooses its allocation of data-processing capacity from $\mathcal{M}$ to maximize expected utility at the trading stage conditional on its information at the data-choice stage, taking the equilibrium price-information environment and the subsequent optimal portfolio choice as given;

(ii) For every investor l of type $L \in \mathcal{L}^R$, $\boldsymbol{q}_l^R \in \arg\max_{\boldsymbol{q} \in \mathcal{Q}_L^R} \mathbb{E}\left[-\exp(-\rho \widetilde{w}_l') \mid \mathcal{I}_l\right]$, subject to the budget constraint (1)*;*

(iii) Investors' conditional beliefs are formed from their private information, soft information when available, and the information contained in equilibrium prices;

(iv) The price function clears the market: the aggregate investor holdings of the risky assets equal the exogenous net position $\bar{\boldsymbol{x}}_t + \widetilde{\boldsymbol{x}}$.

III. Market Access and Endogenous Information Allocation

Building on the common environment, this section isolates the market-access mechanism before portfolio constraints are introduced. The market under partial access provides the benchmark. Domestic institutions equalize the marginal value of information across the two data sources, whereas foreign institutions face a distinct cross-asset-learning problem: information about the inaccessible Type-II asset can still improve trading of the accessible Type-I asset when funda-

mentals are correlated. The close- and full-access markets then show how access changes the allocation of a fixed information-processing capacity across assets.

A. Partial access

This section specializes the modelling framework to the partial-access regime where $\mathcal{L} = \{I, J, K\}$, $\mathcal{Q}_I = \mathcal{Q}_J = \mathbb{R}^2$ and $\mathcal{Q}_K = \mathbb{R} \times \{0\}$.

A.1. Portfolio choices

At the trading stage, investor l faces a simple one-shot portfolio choice problem, which has an optimal solution[3]

$$\boldsymbol{q}_l = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}_L \hat{\boldsymbol{\mu}}_l, \tag{3}$$

where $\hat{\boldsymbol{\mu}}_l \equiv \mathbb{E}[\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_l]$ and $\hat{\boldsymbol{\Omega}}_L \equiv \mathbb{V}ar^{-1}(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_l)$ are the conditional mean and precision of excess payoffs.

A.2. Equilibrium price relation

Definition 2 (Cross-precision). *Let $z \equiv \hat{\mathcal{C}}_{Ky}/\hat{\Omega}_{Ky,(\mathrm{II})}$ denote the cross-precision of the foreign posterior, the off-diagonal entry of $\hat{\boldsymbol{\Omega}}_{Ky}$ normalized by its $\{2,2\}$ element.*

Economically, z measures the intensity of foreign *cross-asset learning*: although foreign institutions can trade only Type-I shares, information about Type II can affect their Type-I demand when fundamentals are correlated.

Signal-to-noise ratio The SNR enters the market clearing condition and is related to other variables through

$$\boldsymbol{\zeta}_t = -\frac{1}{\rho r}(r\boldsymbol{I} - \boldsymbol{G})\Big[\Big(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \Big(\boldsymbol{E}_{11} - z\boldsymbol{E}_{12}\Big)\boldsymbol{M}_K\Big)\boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon\Big], \tag{4}$$

where $\lambda_1 = \lambda_I + \lambda_J$, $\mathbb{E}_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $\mathbb{E}_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. An interesting observation is that the SNR matrix is not symmetric. If foreign institutional investors choose to acquire data on type-II

[3]For notational simplicity, this paper adopts the notational style that uses a hat to distinguish conditional moments at the portfolio choice stage (*interim*) with those at the data choice stage (ex ante).

risky asset, the information they obtain after data processing is not directly aggregated into the equilibrium price of type-Ⅱ risky asset since it is not eligible for international trade. However, this information will be partially reflected in the equilibrium price of type-I risky asset, if the two types of assets are correlated, in which case information on type-Ⅱ asset also affects their posterior belief on type-I. This creates informational inefficiency.

Average posterior precision Market clearing aggregates investors' posterior precision into an eligibility-weighted object $\bar{\hat{\boldsymbol{\Omega}}}_y$. Under partial access it need not be symmetric, because it is an equilibrium aggregate rather than an individual Bayesian precision matrix. Foreign entry dilutes type-I soft information because the mass of investors eligible to hold that asset rises from λ_1 to λ_2; it does not mechanically dilute type-Ⅱ soft information, because type-Ⅱ remains closed to foreign investors. Foreign institutions add private information directly when they process type-I data. When the assets are correlated, their type-Ⅱ data can also enter the type-I price through the cross-SNR, thereby changing the common price-information component $\boldsymbol{\Omega}_p$ and, in particular, potentially raising $\hat{\Omega}_{p,(\mathrm{II})}$ through the cross-price signal. Limited investability creates two countervailing forces. Foreign Type-II information can improve the information embedded in prices through cross-asset learning, but the trading restriction limits the precision that foreign institutions can use directly in Type-I demand.

Price relation Let $\boldsymbol{\Delta}_{\text{pa}}$ denote the eligibility correction induced by foreign Type-II ineligibility; it vanishes under full access. In equilibrium, the price coefficients are characterized by

$$\boldsymbol{A} = \frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - \rho r(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Lambda}^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\bar{\boldsymbol{x}}, \tag{5}$$

$$\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}, \tag{6}$$

$$\boldsymbol{C} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\left[\boldsymbol{I} - \bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Omega}_y + \bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Delta}_{\text{pa}}\right], \tag{7}$$

$$\boldsymbol{D} = \boldsymbol{C}\boldsymbol{\zeta}_t^{-1}, \tag{8}$$

where $\bar{\hat{\boldsymbol{\Sigma}}}_y \equiv \bar{\hat{\boldsymbol{\Omega}}}_y^{-1}$. $\boldsymbol{B}$ is the constant discounted dividend growth rate, and together $\boldsymbol{A}$ and $\boldsymbol{B}$ govern the level of the equilibrium price. $\boldsymbol{C}$ is the price sensitivity to dividend innovation, and $\boldsymbol{D}$ regulates the amount of demand shocks contained in the price. Purchased data amount $\boldsymbol{M}_L$, data processing capacity $\bar{M}$, effective data precision $\boldsymbol{\Pi}_e$, relative data-processing efficiency φ, and

the precision of soft information $\mathbf{\Omega}_\epsilon$ all enter the price coefficients via the SNR and the average posterior precision of the dividend innovation.

A.3. Endogenous information production

The central equilibrium choice is how each institutional investor allocates the fixed capacity $\bar{M}$ across the two data sources. Domestic and foreign institutions solve different allocation problems because domestic investors possess soft information while foreign institutions is only granted access to Type-I asset. The two investor problems are characterized in turn.

Domestic information allocation The relevant sufficient statistic for the allocation problem is summarized in the following lemma.

LEMMA 1. *The data choice of domestic institutional investor i at the trading date enters the ex ante utility function only via* $\det(\hat{\mathbf{\Sigma}}_i)$*, where $\hat{\mathbf{\Sigma}}_i$ is the posterior variance of excess payoffs.*

Since the data choice $(M_{i,(\mathrm{I})}, M_{i,(\mathrm{II})})$ only enter the ex ante utility function through the determinant of the posterior variance of payoff $\det(\hat{\mathbf{\Sigma}}_i)$, the data choice problem reduces to allocating data points to maximize the determinant of the posterior precision $\det(\hat{\mathbf{\Omega}}_i)$. The F.O.C. can be written in effective-precision units as $\frac{\hat{\Omega}_{iy,(\mathrm{I})}}{\varphi^2\Pi_{e,(\mathrm{I})}} = \frac{\hat{\Omega}_{iy,(\mathrm{II})}}{\varphi^2\Pi_{e,(\mathrm{II})}}$. Thus domestic institutional investor i equalizes posterior precision measured in units of the effective precision generated by one processed domestic data point. This normalization motivates the following definition.

DEFINITION 3 (Equivalent data amount). *For asset type $N \in \{\mathrm{I}, \mathrm{II}\}$ and access regime a, the equivalent data amount is* $m_{e,(N);a}(M) \equiv (\Omega_{y,(N)} + \hat{\Omega}_{pt,(N);a} + \Omega_{\epsilon,(N)})/(\varphi^2\Pi_{e,(N)}) + M$.

It is the posterior precision for Type-N dividends, expressed in units of the effective precision from one processed domestic data point. Under partial access we suppress the regime tag and write $m_{e,(N)}$. Conditional on the equilibrium price signal, $m_{e,(N);a}(M) = m_{e,(N);a}(0) + M$, so that one additional processed data point raises the corresponding equivalent-data amount by one. This is an individual-choice identity: an atomistic investor takes $\hat{\Omega}_{pt,(N);a}$ as given when reallocating processing capacity. Across equilibrium regimes, price precision is endogenous, so the level of the equivalent-data amount can change even holding M fixed. The interior first-order condition is therefore $m_{e,(\mathrm{I})}(M_{i,(\mathrm{I})}) = m_{e,(\mathrm{II})}(M_{i,(\mathrm{II})})$.

Denote the optimal Type-I data allocation of domestic institutional investor i by M_i^*; the corresponding Type-II allocation is $\bar{M} - M_i^*$. The optimal Type-I allocation is

$$M_i^* = \begin{cases} 0, & \text{if } m_{e,(\mathrm{I})}(0) \geq m_{e,(\mathrm{II})}(\bar{M}), \\ \bar{M}, & \text{if } m_{e,(\mathrm{II})}(0) \geq m_{e,(\mathrm{I})}(\bar{M}), \\ \frac{1}{2}\big[m_{e,(\mathrm{II})}(\bar{M}) - m_{e,(\mathrm{I})}(0)\big], & \text{otherwise.} \end{cases} \tag{9}$$

The two corner conditions compare the equivalent-data amounts at the opposite endpoints of the capacity constraint; otherwise, the unique interior allocation equalizes the two equivalent-data amounts. Thus the domestic allocation follows a water-filling rule: more processing capacity is directed to the asset with the lower initial equivalent-data stock.

Foreign cross-asset learning Under partial access, foreign institutions can trade Type I but cannot trade Type II. Their information problem is nevertheless two-dimensional: because fundamentals are correlated, processing Type-II data can improve the posterior used to trade Type I. Thus information can have private trading value even when the underlying asset is not directly tradable. The following lemma identifies the sufficient statistic for this information choice.

Lemma 2 (Foreign data choice under partial access). *Conditional on the equilibrium price signal, a foreign institution allocates data to maximize the Type-I Schur-complement precision*

$$\hat{\Omega}^S_{Ky,(\mathrm{I})} = \hat{\Omega}_{Ky,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^2_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}.$$

Thus the Type-II trading restriction does not make Type-II information irrelevant: Type-II data can raise the effective precision used in the foreign investor's Type-I trading problem through the Schur complement. To proceed, define the marginal value of this cross-asset learning.

Definition 4 (Information spillover rate). *Conditional on the equilibrium price signal, let* $M \equiv M_{k,(\mathrm{II})}$ *and write*

$$z(M) \equiv \frac{\mathcal{C}_y + \hat{\mathcal{C}}_p}{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M\Pi_{e,(\mathrm{II})}}.$$

The information spillover rate is

$$\kappa(M) \equiv z(M)^2 \frac{\Pi_{e,(\mathrm{II})}}{\Pi_{e,(\mathrm{I})}}.$$

It is the marginal Type-I precision generated indirectly by one processed Type-II observation, relative to that generated directly by one processed Type-I observation. The optimal data allocation profile therefore depends on the information spillover rate. If $\kappa(0) \leq 1$, the marginal indirect Type-I precision from Type-II processing is already no greater than the direct Type-I precision at the Type-I corner, so the foreign investor allocates no capacity to Type II, $M^*_{k,(\text{II})} = 0$. At the opposite boundary, if $\kappa(\bar{M}) \geq 1$, the indirect marginal precision remains at least as large even after all capacity is assigned to Type II, so $M^*_{k,(\text{II})} = \bar{M}$. Otherwise the unique interior allocation equates the two marginal precision gains. The data choice can be expressed as $(\bar{M} - M^*_{k,(\text{II})}, M^*_{k,(\text{II})})$, where

$$M^*_{k,(\text{II})} = \begin{cases} 0, & \kappa(0) \leq 1, \\ \bar{M}, & \kappa(\bar{M}) \geq 1, \\ M^{\text{int}}, & \kappa(0) > 1 > \kappa(\bar{M}). \end{cases} \tag{10}$$

Here, $M^{\text{int}} \in (0, \bar{M})$ is the unique solution to $\kappa(M^{\text{int}}) = 1$. At an interior allocation, this condition means that the marginal value of direct Type-I learning equals the marginal value of indirect Type-I learning through Type-II information.

Proposition 1 (Information allocation under heterogeneous market access). *Conditional on the equilibrium price-information environment, investors allocate processing capacity according to the marginal value of information. Domestic institutions allocate capacity across assets by comparing their equivalent-data stocks. Under partial access, foreign institutions may optimally process information about an asset they cannot hold whenever the cross-asset information value exceeds the marginal value of direct learning. At an interior foreign allocation, $\kappa(M^*_{k,(\text{II})}) = 1$, so the marginal value of direct Type-I learning equals the marginal value of cross-asset learning from Type-II information. Therefore, investability and information relevance need not coincide.*

Closed-form solutions and the full corner/interior/corner allocation geometry are provided in the Internet Appendix.

While foreign institutional investor k could optimally choose to learn about type-II risky asset, the private value of doing so need not coincide with its social value. Whether this choice constitutes excessive information production is determined by the constrained-planner comparison in Section V; it cannot be inferred from the foreign investor's private problem alone. One implication that does follow directly from the private problem is that cross-asset learning requires fundamental dependence across the two asset types. When the Type-I and Type-II dividend in-

novations are uncorrelated, learning about Type-II does not improve the foreign investor's belief about the tradable Type-I asset.

Thus foreign investors process information about the nontradable asset only when its fundamentals contain information about the tradable asset. For correlated assets, the foreign Type-II data choice varies with effective data precision. At the baseline $|\gamma| \simeq 0.64$, the corrected numerical solution remains at the Type-I corner over the reported technology range.

The formal comparative static treats effective Type-II precision as an equilibrium object: information choices and the price signal adjust jointly.

Corollary 1 (Cross-asset information transmission). *Under partial access, an asset that investors cannot directly hold can still affect information acquisition and equilibrium prices when information about that asset has positive cross-asset value. Hence non-investability does not imply information irrelevance.*

Proposition 2 (Intermediate-precision cross-asset learning). *Maintain $C_y < 0$, and let $E \equiv \Pi_{e,(\mathrm{II})}$ vary through primitive Type-II raw-data precision while β and the remaining primitives are held fixed. Then foreign Type-II processing vanishes at both precision extremes: there exist $0 < \underline{E} < \overline{E} < \infty$ such that*

$$M^*_{k,(\mathrm{II})}(E) = 0$$

for all $0 < E < \underline{E}$ and all $E > \overline{E}$.

If, in addition, a primitive sufficient condition holds, then there exist finite thresholds $0 < E_- < E_+ < \infty$ such that

$$M^*_{k,(\mathrm{II})}(E) > 0 \qquad \text{for every } E \in (E_-, E_+).$$

Hence foreign cross-asset learning is active on a nonempty bounded intermediate-precision region and is inactive for sufficiently low and sufficiently high effective Type-II precision.

At low Type-II precision, indirect learning is unproductive; at high precision, its information is increasingly revealed through domestic learning and prices. These distinct forces generate the intermediate region in Proposition 2. Formal bounds, sufficient conditions, and numerical illustrations are provided in the Internet Appendix.

B. Effects of Partial Access

Market access affects prices through the SNR, average posterior precision, and risk bearing. These effects are the joint outcome of two frictions: the technology–soft-information gap determines the information each investor type brings to the market, while the investability restriction determines where foreign processing capacity can be deployed and incorporated into prices. Foreign entry adds processed information and changes the density of domestic soft information; domestic institutions then reoptimize their allocation of data-processing capacity across the two asset types.

Average posterior precision is summarized by the asset-specific diagonal entries $\bar{\hat{\Omega}}_{y,(N)}$ of the eligibility-weighted precision aggregate $\bar{\hat{\mathbf{\Omega}}}_y$. Foreign entry adds effective hard information, induces domestic institutions to reallocate data acquisition, and changes the SNRs that enter every investor's posterior precision.

Finally, the unit demand level and average posterior variance determine the risk premium embedded in prices. Foreign entry enlarges the pool of investors who can bear risk in eligible shares, consistent with the risk-sharing component of liberalization emphasized in the literature (Chari and Henry, 2004; Chan and Kwok, 2017).

B.1. Domestic information reallocation

The closed-access counterfactual is readily available from the partial-access model by setting $\lambda_K = 0$. Using Definition 3, $m_{e,(N)}$ without a regime tag denotes the equivalent data amount evaluated in the partially liberalized equilibrium, while $m_{e,(N);\text{cl}}$ denotes the same normalization evaluated in the closed-access equilibrium ($\lambda_K = 0$). Accordingly, $\hat{\Omega}_{pt,(N);\text{cl}}$ is the closed-equilibrium price-signal precision with all foreign-investor terms set to zero. These are regime evaluations of a single concept rather than separate definitions; their levels can differ because equilibrium price precision changes with market access.

B.2. Price informativeness and the signal-to-noise ratio

Partial market access affects the SNRs of type-I and type-II risky assets. Beyond the common channel through changes in domestic institutional investors' data choices, which affects both asset types, the effects differ across assets.

For type-I risky assets, foreign entry can affect the SNR directly or indirectly. If foreign institutions learn about type-I assets, the information extracted from the acquired and processed data is incorporated into the Type-I equilibrium price, directly affecting the Type-I SNR. The effect increases with the foreign investor mass, processing capacity, and data-signal precision. If foreign institutions instead learn about type-II assets, the resulting information is not incorporated directly into the Type-II equilibrium price; it affects the Type-I price only through information spillovers and therefore affects the Type-I SNR indirectly.

For type-II risky assets, domestic institutional investors are the sole contributors to the SNR. Since market access does not affect their endowed soft information, the direction of the SNR change depends only on domestic institutions' data choices: if they learn less about type-II assets, i.e. $M_{(\mathrm{II})} < M_{(\mathrm{II});\mathrm{cl}}$, the Type-II SNR falls; if they learn more, it rises.

Let us call $\zeta_{(\mathrm{I})}$ and $\zeta_{(\mathrm{II})}$ the own-asset SNRs, which measure the loading of each asset's own dividend-innovation signal relative to noise in its equilibrium price, and $\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}$ the cross-asset SNR, which measures the loading of type-II dividend information into the type-I price relative to type-I noise. These ratios depend on the risk-free rate, dividend growth, risk aversion, investor masses, effective data precision $\mathbf{\Pi}_e$, relative processing efficiency φ, and institutional data choices.

Notice that $\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}$ is present only if foreign institutional investors choose to learn about type-II assets. Under the type-II corner $\kappa(\bar{M}) \geq 1$, foreign institutions allocate all processing capacity to type-II data. Their information therefore contributes to the cross-asset SNR rather than directly to either own-asset SNR. Any change in the two own-asset SNRs is then driven by domestic institutions' reallocation of their fixed processing capacity across the two asset types.

Which friction drives the result. Result (1) is driven mainly by the second mismatch (investability): foreign hard information is aggregated into the type-1 price, and the positive effect is governed by the foreign presence and technology relative to domestic institutions ($\lambda_K > \lambda_I \varphi^2$). The negative case in (2) requires the first mismatch's channel—a reduction in domestic learning about type-1—to be sufficiently large; the soft-information dilution term $\lambda_1 \Omega_\epsilon$ enters the threshold.

At this all-Type-II foreign-processing corner, the changes normalized by the asset-specific mapping from effective data precision into price informativeness are exactly equal and opposite:

$$\frac{\Delta\left|\zeta_{(\mathrm{I})}\right|}{(r-g_1)\Pi_{e,(\mathrm{I})}} = -\frac{\Delta\left|\zeta_{(\mathrm{II})}\right|}{(r-g_2)\Pi_{e,(\mathrm{II})}}. \tag{11}$$

Thus, at this corner, if domestic institutions increase type-I processing after partial access, the type-I own-asset SNR rises and the type-II own-asset SNR falls; if they decrease type-I processing, the signs reverse. If domestic institutions do not reallocate processing capacity, both own-asset SNR changes are zero. The raw own-asset SNR changes need not be equal in magnitude because the two asset types can differ in effective data precision and dividend-growth exposure.

Which friction drives the result. The opposite-sign movement comes directly from domestic institutions' fixed processing-capacity constraint: more processing of one asset mechanically means less processing of the other. The investability mismatch instead explains why foreign institutions are at the type-II corner and why their type-II information enters price informativeness through the cross-asset SNR rather than directly through either own-asset SNR. Equation (11) removes the asset-specific scaling by $(r - g_N)\Pi_{e,(N)}$ and makes the one-for-one domestic reallocation transparent.

The corollary emphasizes that making type-II data investable does not mechanically increase its price informativeness. Foreign institutional investors bring additional effective precision, but this gain can be offset when entry crowds out domestic type-II information production. The exact condition is therefore a comparison between lost domestic effective information and newly created foreign effective information rather than a threshold on foreign investor mass alone.

B.3. Total information and posterior precision

Foreign entry changes total information through three channels: it adds foreign hard information, changes domestic data acquisition, and changes the density of domestic soft information among eligible investors. The last channel mechanically affects Type I but not Type II, because foreign investors become eligible only for Type-I shares under partial access.

The net effect on posterior precision is therefore not determined by price informativeness alone. At the all-Type-I foreign-processing corner, a weak decline in Type-I price informativeness implies a strict decline in average Type-I posterior precision because foreign entry can both reduce domestic information density and impose the portfolio-constraint discount. For Type II, the eligible-investor mass is unchanged, and the cross-price signal can offset a decline in domestic Type-II information production.

C. Closed access

In the closed economy, the domestic financial market remains closed to foreign investors at the trading date, so that only domestic investors trade the two risky securities. The environment is the specialization of Section II in which foreign investor masses are set to zero while the domestic primitives are held fixed. This makes closed access the natural theoretical benchmark for isolating the effect of foreign entry and asset eligibility.

The information problem of the closed economy is the domestic part of the Environment and Market Access: domestic institutional investor i chooses $(M_{i,(\mathrm{I})}, M_{i,(\mathrm{II})})$ subject to the data-processing capacity $\bar{M}$ before observing prices, domestic individual investors j trade on price and soft information only, and the equilibrium price aggregates the private information of domestic institutional investors. With no foreign investors, there is no foreign demand, no foreign data processing, and no dilution of the domestic soft-information density; the price signal $\boldsymbol{s}_p$ is generated by domestic investors alone. Formally, the closed equilibrium specializes Definition 1 to $\mathcal{L}_2 = \{I, J\}$ (equivalently, the $\lambda_K = 0$ case): the equilibrium price function, data strategies, and portfolio choices satisfy the same conditions with all foreign-investor terms set to zero.

D. Full access

Instead of a partial market access, economies theoretically could also directly enter a full market access. The baseline full market access model in this section on the one hand is useful in its own regard, and on the other hand provides a benchmark and long-term vision for partially liberalized markets.

In a fully liberalized economy, all securities are available for international trade. In addition, λ_H foreign individual investors also enter the domestic market. Foreign institutional investors might allocate data processing to both types of risky assets. As a result, information will be less concentrated on a specific type of risky asset. The entry of foreign individual investors also lowers the information density.

The full-access equilibrium specializes Definition 1 to $\mathcal{L} = \{I, J, K, H\}$, with $\mathcal{Q}_L = \mathbb{R}^2$ for every investor type.

D.1. Full-access equilibrium

The optimal portfolio choices have the same form as in the partial-access model. The optimal data choice problem of domestic institutional investors does not change, but that of foreign institutional investors is no longer trivial because they can now also invest in the type-II risky asset. They therefore allocate $\bar{M}$ data-processing capacity across both types of risky assets, as domestic institutional investors do. The form of the price coefficients changes with the set of investor types allowed to hold each risky asset.

Under full access both assets have the same eligible-investor mass $\lambda_3 \equiv \lambda_I + \lambda_J + \lambda_K + \lambda_H$. For $N \in \{\mathrm{I}, \mathrm{II}\}$, the own-asset SNR and the asset-specific average posterior precision are

$$\zeta_{(N);\mathrm{fu}} = -\frac{r - g_N}{\rho r}\left[(\lambda_I \varphi^2 M_{I,(N);\mathrm{fu}} + \lambda_K M_{K,(N);\mathrm{fu}})\Pi_{e,(N)} + \lambda_1 \Omega_{\epsilon,(N)}\right], \tag{12}$$

$$\bar{\hat{\Omega}}_{y,(N);\mathrm{fu}} = \Omega_{y,(N)} + \hat{\Omega}_{p,(N);\mathrm{fu}} + \frac{\lambda_I \varphi^2 M_{I,(N);\mathrm{fu}} + \lambda_K M_{K,(N);\mathrm{fu}}}{\lambda_3}\Pi_{e,(N)} + \frac{\lambda_1}{\lambda_3}\Omega_{\epsilon,(N)}, \tag{13}$$

where $\hat{\Omega}_{p,(N);\mathrm{fu}} = \zeta^2_{(N);\mathrm{fu}}\Omega_{x,(N)}$. Unlike partial access, no portfolio-constraint Schur discount appears because both risky assets are tradable by foreign institutions.

D.2. Comparative statics of full market access

Full market access influences the SNRs in a slightly different way. What foreign institutional investors have learned about type-II risky asset, if any, is aggregated by the market. Domestic institutional investors may change their data acquisition accordingly. The change in average posterior precision from partial to full access contains both a price-information component and a change in the density of privately held information.

Proposition 3 (Market access and equilibrium price informativeness). *Market access need not move price informativeness in the same direction as private or posterior information.*

(i) Making Type-II shares newly investable can reduce their price informativeness when the loss of cross-price information exceeds the gain in their own-price information.

(ii) Under partial access, Type-I price informativeness can rise even while the average posterior precision of eligible investors falls.

Hence price informativeness is an equilibrium aggregation outcome and need not be monotone in either investability or investors' private information.

The exact conditions and proofs are collected in Appendix A.

As China has gradually broadened and deepened its stock-market liberalization, some researchers have found evidence that stock prices in China have become as informative about future profits as those in the United States (Carpenter et al., 2021).

IV. Portfolio Constraints and Endogenous Participation

The preceding section isolates how market access changes information allocation and aggregation. This section adds the short-sale constraint, which introduces an additional margin: investors need not hold every risky asset available to them. Information therefore affects not only posterior precision but also the probability that an investor participates in each asset market. These participation decisions change aggregate risk-bearing capacity and, through market clearing, the information content of prices. The Internet Appendix provides the KKT derivations, full fixed-point system, regularity conditions, and full-access specialization.

A. Portfolio Choice and Market Participation

With a short-sale constraint, a domestic investor need not hold both risky assets. Depending on her posterior expected excess payoffs, she may hold both assets, only Type I, only Type II, or neither. We denote these four portfolio states by 11, 10, 01, and 00, respectively. Because the two assets are correlated, the decision to hold one asset generally depends on the expected payoff of the other. Hence the constrained portfolio is not obtained simply by setting negative components of the unconstrained portfolio equal to zero: when one asset is excluded, the remaining position is reoptimized.

Information affects constrained demand through two channels: it changes the amount of risk an active investor wishes to bear and the probability that the investor participates in each asset market.

B. Partial access with a short-sale constraint

Under partial access, domestic positions are nonnegative and foreigners may hold only nonnegative Type-I positions; the Type-II zero for foreigners is an investability restriction rather than a binding short-sale constraint. A foreign investor's effective Type-I precision remains the Schur

complement, but information now also affects the probability of taking a positive Type-I position. Thus the constraint changes the private value of information as well as risk bearing.

Exact constrained aggregation makes the global price map nonlinear. We therefore use an equilibrium that is exact in portfolio choice and aggregation and locally linear only in the price-information map around the reference state.

C. Closed and full access with a short-sale constraint

The same participation mechanism applies under closed and full access. Closed access isolates domestic participation; partial access adds foreign participation in Type I only; and full access makes foreign participation in both asset markets endogenous.

D. Relation to the Unrestricted Model

As the short-sale constraint becomes slack, participation converges to one on every eligible asset and the constrained economy converges to the unrestricted benchmark.

V. Welfare Analysis

Having characterized decentralized information acquisition, this section compares it with the allocation chosen by a planner and identifies the wedge between private and social marginal values.

The cross-asset learning mechanism does not by itself establish a welfare loss. To evaluate efficiency, consider a constrained second-best planner that chooses only the common data allocation of each institutional-investor type. The planner preserves market access, short-sale constraints, private signals, decentralized portfolio choice, and rational-expectations market clearing. The Internet Appendix provides the complete welfare derivation and formulation.

Let a denote the access tag, with cl for closed access, no access tag for the partially open benchmark, and fu for full access. Because this section preserves the short-sale constraint, generic decentralized welfare objects carry the additional tag c and planner allocations append pl. Thus $\mathrm{CE}_{L;a,\mathrm{c}}(\boldsymbol{m})$ is type L's ex ante certainty equivalent in the constrained decentralized equilibrium under access regime a, and the corresponding planner allocation carries $a, \mathrm{c}, \mathrm{pl}$. The vector $\boldsymbol{m}$

collects institutional data allocations. The global planner and domestic regulator solve, respectively,

$$\boldsymbol{m}_{G;a,\mathrm{c},\mathrm{pl}} \in \arg\max_{\boldsymbol{m}} \sum_{L\in\mathcal{L}_a} \lambda_L \,\mathrm{CE}_{L;a,\mathrm{c}}(\boldsymbol{m}), \qquad \boldsymbol{m}_{D;a,\mathrm{c},\mathrm{pl}} \in \arg\max_{\boldsymbol{m}} \left[\lambda_I \,\mathrm{CE}_{I;a,\mathrm{c}}(\boldsymbol{m}) + \lambda_J \,\mathrm{CE}_{J;a,\mathrm{c}}(\boldsymbol{m})\right], \tag{14}$$

subject to the same capacity, portfolio, price-consistency, and market-clearing conditions as the decentralized economy. In the main benchmark, information-processing capacity is an exogenously committed budget: for every data-producing type L, $M_{L,(\mathrm{I})} + M_{L,(\mathrm{II})} = \bar{M}$. Thus the planner can reallocate capacity across the two assets but cannot leave it idle. This restriction preserves the paper's allocation problem and isolates whether decentralized investors direct a given processing budget to the socially efficient source.

An atomistic investor takes the price law as given when choosing data. The planner additionally internalizes how a common change in type R's data allocation changes prices, posterior moments, and the probability that trading constraints bind. Denote this omitted marginal equilibrium effect by $\mathcal{X}_{R;a,\mathrm{c}}$. At an interior decentralized allocation, the private marginal value is zero, so

$$\left.\frac{dW_{a,\mathrm{c}}}{dm_R}\right|_{\boldsymbol{m}_{a,\mathrm{c}}} = \mathcal{X}_{R;a,\mathrm{c}}(\boldsymbol{m}_{a,\mathrm{c}}). \tag{15}$$

Consequently, a nonzero $\mathcal{X}_{R;a}$ establishes that decentralized information production is not constrained efficient.

For the partial-opening mechanism, let $n_K = M_{K,(\mathrm{II})}$ be the foreign information-capacity choice assigned to the inaccessible asset. Its decentralized and planner equilibrium values under partial access are $n_{K;\mathrm{c}}$ and $n_{K;\mathrm{c},\mathrm{pl}}$. If $-\mathcal{X}_{K;\mathrm{c}}(\boldsymbol{m}_{\mathrm{c}}) < 0$, a marginal reduction in n_K raises welfare; under strict concavity, $n_{K;\mathrm{c},\mathrm{pl}} < n_{K;\mathrm{c}}$. The fixed-active-set benchmark makes this condition explicit. If c is posterior cross-precision, τ_{I} and τ_{II} are the two signal precisions per observation, and $0 \le \chi_{\mathrm{c}} \le 1$ is the endogenous social pass-through of the private cross-learning benefit, then, when both choices are interior,

$$n_{K;\mathrm{c}} - n_{K;\mathrm{c},\mathrm{pl}} = \frac{|c|\left(1-\sqrt{\chi_{\mathrm{c}}}\right)}{\sqrt{\tau_{\mathrm{I}}\tau_{\mathrm{II}}}}. \tag{16}$$

The wedge is positive when correlation is nonzero and $\chi_{\mathrm{c}} < 1$, vanishes when correlation is zero, and shrinks when broader access raises social pass-through.

Equations (15)–(16) therefore provide conditional analytical predictions rather than a calibrated sign restriction. The quantitative section evaluates the decentralized and planner allocations under the estimated primitives and reports the associated certainty-equivalent welfare gains separately.

VI. Quantitative Analysis

This section takes the model to the Shanghai–Hong Kong Stock Connect setting. It first maps the institutional environment into the calibration, then evaluates the benchmark, policy counterfactuals, and reduced-form evidence. More broadly, the framework applies to settings in which investors differ in market access, information acquisition, and trading constraints.

A. Institutional setting

The domestic Chinese stock market (the "A-share" market) started in 1990 with the establishment of two independently operated domestic stock exchanges: the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE). Historically, the Chinese stock market has been only weakly integrated with international markets. The Shanghai–Hong Kong Stock Connect Program was an important step in China's capital-market liberalization. Launched on November 17, 2014, it allows local and international investors in Hong Kong to trade SSE shares (northbound trading) through the Hong Kong Stock Exchange (HKEX). Shanghai–Hong Kong Stock Connect primarily covers large-capitalization index constituents listed on the SSE and HKEX. The Internet Appendix provides the trading arrangement, eligibility criteria, and international comparisons.

B. Parameterization and Estimation

The calibration proceeds in three steps. First, the investor-type shares and the risk-free rate are calculated directly from the data. Second, the price coefficients are estimated by OLS from cross-sectional regressions. The estimated matrix $\widehat{\boldsymbol{C}}$ provides the price-loading moment residuals, while $\widehat{\boldsymbol{B}}$ is used entirely in a preliminary step to recover the dividend-growth parameters $\boldsymbol{G}$ through $\boldsymbol{G} = r\widehat{\boldsymbol{B}}(\boldsymbol{I} + \widehat{\boldsymbol{B}})^{-1}$; with $\boldsymbol{G}$ fixed, the model-implied $\boldsymbol{B}$ coincides with $\widehat{\boldsymbol{B}}$ by construction, so $\widehat{\boldsymbol{B}}$ residuals carry no identifying information and are excluded from the third-stage objective. Third, the remaining structural parameters are estimated by weighted minimum distance,

minimizing the price-loading ($\widehat{C}$), price-noise-covariance, ownership-share, and holdings-ratio residuals, all evaluated at the fully constrained local equilibrium with the data choices solved endogenously as an inner fixed point and the short-sale constraints imposed asset by asset.

Identification The different sets of moments identify distinct components of the model. Ownership-share moments identify the relative investor masses $\boldsymbol{\lambda}$ after accounting for equilibrium portfolio demand. Price-loading moments identify the covariance structure of information and noise, because endogenous information acquisition affects equilibrium price sensitivities. Price-noise moments discipline the supply-noise parameters conditional on the equilibrium price signal. The foreign cross-asset precision object z is an endogenous equilibrium outcome rather than an additional data moment; the spillover channel is disciplined through the price coefficients and the equilibrium structure.

B.1. Measured inputs and calibration targets

The observed investor-type shares and the risk-free rate are measured inputs rather than estimated parameters: the shares are calibration targets for the investor masses $\boldsymbol{\lambda}$, and r is a fixed input. The scale of the information technology is set by three further choices: $\beta_I = 1$ and $\bar{M} = 2$ are normalizations, whereas $\beta = 3$ (equivalently $\varphi = \beta_I/\beta = 1/3$) is a substantive fixed technology assumption. The supplementary analyst-forecast exercise is informative against equal processing efficiency but does not sharply identify the magnitude of the gap: joint or conditional re-estimation improves fit only by moving risk aversion to an economically extreme region. We therefore retain $\beta = 3$ as the maintained benchmark and report the forecast construction, profiles, and joint estimates in the Internet Appendix.

Market share The aggregate data on the composition of shareholders are obtained from China Securities Regulatory Commission (CSRC). The largest holder of Chinese A-shares is industrial capital, including non-financial companies and general legal persons. Considering that industry capital and original shareholders' holdings cannot be freely circulated[4], they are not included in the analysis. In the same vein, market shares are calculated with *negotiable market capitalization*, i.e. market capitalization for tradable common listed shares only.

[4]These are common shares held by state legal persons or original promoters. They cannot be traded on the exchange but can be traded by auction or transfer by agreement under the approval of the CSRC.

In 2013, individual investors held more than 33% of total A-share market capitalization, peaking above 39% in 2015 and declining below 30% by 2023; domestic institutional investors held around 15%, and foreign capital about 3%. These ratios are computed against total market capitalization, which includes non-tradable shares predominantly held by industrial capital and original shareholders. Since only negotiable market capitalization is used in this model, these shares are restated on the negotiable basis: for the time window surrounding the launch of the Shanghai-Hong Kong Stock Connect, domestic individual investors, domestic institutional investors, and foreign institutional investors hold 66%, 28%, and 6% of negotiable market capitalization, respectively, which sum to one.

Risk-free rate The risk-free rate is set to the gross return $r = 1.042$, corresponding to a mean yield to maturity of approximately 4.2% per annum on the 10-year government bond in 2014. The government bond yield curve is published by China Central Depository & Clearing Co., Ltd. (CCDC).

B.2. OLS estimation for price coefficients

Data on daily stock prices and the eligibility details of the Stock Connect Programs are obtained from the China Stock Market & Accounting Research (CSMAR) database. Following the measurement methodology in the literature, unexpected earnings proxy for the dividend innovation $\widetilde{y}$. Earnings surprises are proxied by residuals from the corresponding time-series forecasting models. To estimate the price coefficients, we run the following cross-sectional regressions:

$$ma_{i,t,(\mathrm{I})} = \hat{A}_{t,(\mathrm{I})} + \hat{B}_{t,(\mathrm{I})} ea_{i,t-1,(\mathrm{I})} + \hat{C}_{t,(\mathrm{I})} ue_{i,t,(\mathrm{I})} + \hat{C}^{(c)}_{t} ue_{i,t,(\mathrm{II})} + \hat{K}_{i,t,(\mathrm{I})} Z_{i,t,(\mathrm{I})} + \hat{\varepsilon}_{i,t,(\mathrm{I})},$$
$$ma_{i,t,(\mathrm{II})} = \hat{A}_{t,(\mathrm{II})} + \hat{B}_{t,(\mathrm{II})} ea_{i,t-1,(\mathrm{II})} + \hat{C}_{t,(\mathrm{II})} ue_{i,t,(\mathrm{II})} + \hat{K}_{i,t,(\mathrm{II})} Z_{i,t,(\mathrm{II})} + \hat{\varepsilon}_{i,t,(\mathrm{II})},$$

where $ma_{i,t}$ is the logarithm of market capitalization divided by total assets; $ea_{i,t-1}$ is the previous year's EPS times shares outstanding, divided by total assets; $ue_{i,t}$ is the earnings surprise normalized by total assets; and $Z_{i,t}$ is a collection of dummies at the 2012 CSRC sub-industry level. The Internet Appendix provides detailed variable construction. Because the outcome is the log of the market-to-assets ratio, the estimated coefficients are semi-elasticities of market capitalization with respect to earnings and earnings surprises—the units in which the equilibrium price-loading moments are matched. These regression coefficients provide empirical moments

corresponding to the model's equilibrium price sensitivities $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$, rather than direct estimates of structural coefficients. The residuals are used as an empirical proxy for the price-noise component $\boldsymbol{D}\boldsymbol{\Sigma}_x\boldsymbol{D}'$, in the sense that any omitted fundamental, measurement, or misspecification component orthogonal to the controls is also absorbed into the residual covariance.

The supply-noise precisions $\Omega_{x,(\mathrm{I})}$ and $\Omega_{x,(\mathrm{II})}$ are therefore jointly identified with the other structural parameters under this measurement assumption rather than observed directly; the Internet Appendix reports a robustness exercise that re-weights the price-noise block and leaves the estimates and the allocation corners unchanged; the exercise demonstrates robustness to the block's weight rather than validation of this maintained measurement mapping, because the regression residual covariance also absorbs omitted fundamentals, measurement error, and misspecification.

The growth matrix $\boldsymbol{G}$ is identified directly from the estimated equilibrium price-loading matrix $\widehat{\boldsymbol{B}}$ through the restriction $\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}$. This identification follows from the present-value restriction linking equilibrium prices to expected dividend growth rather than from an independent estimate of future growth rates. The estimates yield $g_1 > g_2$. This ordering is economically plausible given the selection of foreign-investable securities toward large, liquid, and relatively strong-performing firms, although it is not imposed by the model and should not be interpreted as a mechanical consequence of market access itself.

B.3. Weighted minimum-distance estimation

Let $m_L(\boldsymbol{\lambda})$ denote the model-implied market share of type-L investors,

$$m_L(\boldsymbol{\lambda}) \equiv \frac{\lambda_L\, \boldsymbol{q}_L'\boldsymbol{p}}{\sum_{L'} \lambda_{L'}\boldsymbol{q}_{L'}'\boldsymbol{p}}, \qquad L \in \{I, J, K\}. \tag{17}$$

Both price weights and market shares are homogeneous of degree zero in $\boldsymbol{\lambda}$, so it is convenient to normalize $\lambda_I + \lambda_J + \lambda_K = 1$.

Setting the observed market shares of each type of investors directly as the exogenous weights λ_L is problematic. This mapping is exact only when equilibrium demands are identical across types; with information-driven dispersion in demands, the observed market shares are equilibrium objects that also embed the cross-type dispersion of posterior expectations. Therefore $\boldsymbol{\lambda} = (\lambda_I, \lambda_J, 1 - \lambda_I - \lambda_J)$ is to be estimated by moment matching.

Because price informativeness and posterior variances depend on endogenous data choices,

each candidate parameter vector is evaluated at the fixed point of the constrained partial-access equilibrium. The calibration matches price loadings, price-noise covariance, three investor ownership shares, and the domestic institutional holdings ratio. Effective data precision $\mathbf{\Pi}_e$ is estimated directly, while $\beta = 3$, $\beta_I = 1$, and $\bar{M} = 2$ are maintained benchmark choices.

The Type-II effective-precision parameter is set-identified at the estimated corner because no investor processes Type-II data there; we report a corner-supporting evaluation point and do not interpret its level as a point-identified economic fact. The holdings-ratio moment disciplines the domestic allocation through equilibrium restrictions. The full moment vector, weighting, local-rank diagnostics, and joint-estimation robustness exercises for relative processing efficiency are in the Internet Appendix. Table I reports the estimated primitives and fit. Ownership shares are equilibrium outcomes rather than investor-mass parameters, and institutional information choices are solved endogenously within estimation.

Calibration results The parameter estimates, targeted moments, endogenous information allocation, and participation outcomes are summarized in Tables I and II. As an untargeted diagnostic, the estimated model reproduces the direction of the data-based price variance reduction: 9.32% versus 12.37% in the data for type-I shares and 4.15% versus 4.40% for type-II shares. Because this moment is not separately targeted in estimation, the approximate agreement is reassuring; it is not, however, an independent out-of-sample validation, since the diagnostic is built from the same price-information objects that underlie the targeted $\boldsymbol{C}$ and price-noise moments. The residual tension in Table I now concentrates in the price-noise moments: the model's $\Sigma_{11}^{p,\text{noise}}$ is 0.3008 against data of 0.23479 (marginal residual 3.00) and $\Sigma_{22}^{p,\text{noise}}$ is 0.2257 against 0.32837 (marginal residual -3.51), while the own-price loadings fit well, with C_{22} at 2.2874 against 2.2683 (marginal residual 0.07). The reparameterized estimation moves the residual tension from the non-investable own-price loading toward the price-noise covariance; we leave a fuller account of this trade-off to the Internet Appendix sensitivity discussion rather than disguising it.Reweighting the price-noise block at half its sampling weight and re-estimating leaves every parameter estimate and the allocation corners unchanged (Internet Appendix), so the imperfect fit of these two moments does not drive the estimated information allocation. A related point concerns the cross-loading: the reduced-form cross-price coefficient is positive ($C_{12}^{\text{data}} = 1.1766$), yet the estimated structural equilibrium generates a much smaller cross-loading ($C_{12}^{\text{model}} = 0.0632$). This is not a statistical failure (marginal residual -1.05), and it is economically coherent: investors

internalize correlated fundamentals through private information acquisition, so the incremental informational content of prices for the cross fundamental is small even though the reduced-form comovement—which also reflects discounting and equilibrium trading—is sizable.

Table I
Structural parameter estimates and benchmark model fit.

Panel A reports the primitives of the estimated partial-access economy with short-sale constraints. $\beta = 3$ is maintained; β_I and $\bar{M}$ are normalizations. $\Pi_{e,(\mathrm{II})}$ is a corner-supporting, set-identified value; the Internet Appendix characterizes its identified region. Growth rates are recovered from $\widehat{\boldsymbol{B}}$, and underlying data precision is $\Omega_{e,(n)} = \Pi_{e,(n)}/\beta^2$.

Panel A. Structural parameters

Parameter	Economic interpretation	Value	Status
$\Pi_{e,(\mathrm{I})}$	Effective data precision, type I	7,438.0460	Estimated
$\Pi_{e,(\mathrm{II})}$	Effective data precision, type II	73.8648	Set-identified (corner-supporting)
$\Omega_{\epsilon,(\mathrm{I})}$	Domestic soft-information precision, type I	1,596.7535	Estimated
$\Omega_{\epsilon,(\mathrm{II})}$	Domestic soft-information precision, type II	1,108.4729	Estimated
$\Omega_{x,(\mathrm{I})}$	Supply-noise precision, type I	0.007692	Estimated
$\Omega_{x,(\mathrm{II})}$	Supply-noise precision, type II	0.007692	Estimated
ρ	Risk aversion	1.4613	Estimated
β/β_I	Foreign/Domestic data processing technology	3.0000	Fixed
λ_I	Domestic institutional investor mass	0.2320	Estimated
λ_J	Domestic individual investor mass	0.7485	Estimated
λ_K	Foreign institutional investor mass	0.0195	Implied

C. Benchmark Equilibrium and Information Mechanism

In the calibrated model, foreign institutional investors hold a 6.02% share of negotiable market value (Table I), close to the 6% observed in the data, yet partial access generates equilibrium effects on both the investable and the non-investable shares relative to the closed-market counterfactual

Table I

Structural parameter estimates and benchmark model fit (continued).

The criterion is weighted minimum distance, not efficient GMM: price-loading and price-noise residuals have unit weight, ownership shares use a ten-percentage-point scale, and the holdings ratio enters in logs with unit scale. Marginal residuals use their own standard error or scale. The fit balances the moment blocks rather than exactly matching every moment.

Panel B. Targeted moments and benchmark fit

Moment	Data	Model	Sampling s.e./scale	Marginal residual
C_{11}	3.7123	3.6565	1.0374	−0.05
C_{12}	1.1766	0.0632	1.0610	−1.05
C_{22}	2.2683	2.2874	0.2753	0.07
$\Sigma_{11}^{p,\text{noise}}$	0.23479	0.3008	0.02201	3.00
$\Sigma_{12}^{p,\text{noise}}$	0.00007	0.0040	0.01718	0.23
$\Sigma_{22}^{p,\text{noise}}$	0.32837	0.2257	0.02923	−3.51
Domestic institutional share	0.2800	0.2811	0.10	0.01
Domestic individual share	0.6600	0.6587	0.10	−0.01
Foreign institutional share	0.0600	0.0602	0.10	0.00
Domestic inst. Type-I/II holdings ratio	1.5953	1.6232	1.0	0.02
Minimum-distance objective		1.2587		

Table II
Benchmark equilibrium and information mechanism.

Values refer to the estimated partial-access equilibrium with short-sale constraints. Panels report ownership and participation, information technology and capacity allocation, zero-holding probabilities, and variance reduction relative to the prior. Capacity is measured relative to $\bar{M} = 2$; the active foreign mass equals λ_K times the Type-I participation probability.

	Benchmark value
Panel A. Ownership, mass, and participation	
Foreign institutional ownership share	6.02%
Estimated foreign institutional model mass, λ_K	1.95%
Probability of a zero type-I holding	92.8%
Type-I participation probability	7.2%
Participating foreign mass, $\lambda_K \times 7.2\%$	0.14%
Panel B. Information technology and allocation	
Foreign processing technology, β	3.000
Domestic institutional processing technology, β_I	1.000
Relative domestic processing efficiency, $\varphi \equiv \beta_I/\beta$	1/3
Effective precision per processed observation, foreign relative to domestic	9×
Domestic capacity allocated to Type I / Type II	2.000 / 0.000
Foreign capacity allocated to Type I / Type II	2.000 / 0.000
Panel C. Probability of a zero holding	
Domestic institutions, I: Type I / Type II	77.7% / 73.7%
Domestic individuals, J: Type I / Type II	72.8% / 73.7%
Foreign institutions, K: Type I	92.8%
Panel D. Market outcomes	
Price-based variance reduction, Type I / Type II	9.32% / 4.15%
Total variance reduction, Type I / Type II	83.97% / 67.23%

(Table III). Table II collects the benchmark facts about the foreign sector that make this possible. The central fact is that the foreign sector is quantitatively small along every margin: its estimated investor mass is only $\lambda_K = 0.0195$ (1.95% of the investor population), and its endogenous type-I participation probability is just 7.2%, so the actively participating foreign mass is approximately 0.14% of the population—yet the equilibrium effects of foreign entry are not proportional to any of these margins. A small observed ownership share therefore does not imply a proportionally small equilibrium role. The reason is that ownership share is not the correct sufficient statistic for equilibrium influence: foreign investors differ from domestic investors in information-processing technology, information endowments, asset access, and endogenous participation, and their information and trades enter the public price signal, which feeds back into the decisions of the much larger domestic sector.

The first reason is that observed ownership is an equilibrium outcome, not the foreign investor mass. Market shares are determined by equilibrium holdings (17): a low observed share can reflect a high probability of a zero holding induced by portfolio constraints, smaller positive positions conditional on participation, differences in beliefs and posterior precision, or asset-access restrictions. The calibration estimates a foreign institutional model mass of $\lambda_K = 0.0195$, or 1.95% of the investor population (Table II), even though foreign institutions hold only 6.02% of market value in the benchmark equilibrium. The gap reflects the short-sale constraint, which binds frequently: the type-I zero-holding probability is 92.8% (Table II), so the participation probability is only 7.2%, and the participating foreign mass is approximately $0.0195 \times 0.072 = 0.14\%$. The estimated foreign mass is itself small, and infrequent participation makes the actively trading foreign sector smaller still. Ownership share, investor mass, and informational influence are three distinct objects: even a quantitatively tiny participating foreign mass can exert disproportionate informational influence if its information is sufficiently productive and is transmitted through equilibrium prices. This conclusion is conditional on the benchmark identification region: an almost equally fitting specification with equal type-I and type-II effective precision (objective 1.2919 versus 1.2587) flips the domestic information allocation to type-II and eliminates the type-I foreign-entry effect (Internet Appendix), so the data do not yet globally select this mechanism.

The second reason is that foreign institutions have a processing advantage under the maintained benchmark technology: $\beta = 3$, whereas domestic institutional technology is normalized to $\beta_I = 1$ (Table II), so $\varphi = 1/3$ and $\varphi^2 = 1/9$. The supplementary forecast-error exercise

does not sharply identify this gap: appending the moment can improve fit only by moving risk aversion to an economically extreme region, and the profile is nearly flat over a wide range of processing advantages. We therefore retain $\beta = 3$ as a maintained benchmark and report the forecast construction, profile, and joint-estimation results in the Internet Appendix. With effective data precision $\mathbf{\Pi}_e \equiv \beta^2 \mathbf{\Omega}_e$, one correctly processed observation contributes $\mathbf{\Pi}_e$ to posterior precision when processed by a foreign institution and $\varphi^2 \mathbf{\Pi}_e$ when processed by a domestic institution: holding the data source fixed, a foreign-processed observation delivers nine times the effective precision of the same observation processed with the domestic institutional technology. A marginal foreign investor is therefore informationally different from a marginal domestic holder. In the benchmark, both foreign and domestic institutions allocate their entire processing capacity to type-I data (Table II); this concentration is consistent with the targeted ratio of domestic institutional holdings in type-I to type-II shares of about 1.60 in the data and 1.62 in the model. A small ownership stake can accordingly be associated with a concentrated, high-precision contribution to the information environment of the investable market.

Third, foreign information does not remain private to the foreign sector. Foreign informed demand transmits part of that information into equilibrium prices, and the price is a public signal observed by all investors: foreign entry and information production shift foreign informed demand, which changes the equilibrium price signal, which updates domestic posterior beliefs, which alter domestic information allocation and portfolio demand, and thereby feed back into the price. A foreign sector can therefore have market-wide effects even when its direct holdings are small, because it affects the marginal price signal rather than the average ownership share. This is fundamentally different from a partial-equilibrium ownership-share calculation.

Fourth, domestic institutions do not hold their information production fixed when foreign investors enter. Their processing capacity is allocated endogenously across the two asset types, and a change in price informativeness changes the marginal value of private data. Foreign entry therefore has both a direct information effect, through foreign processing and informed demand, and an indirect crowding-in or crowding-out effect through the reoptimization of domestic information acquisition. The equilibrium effect of foreign entry need not be proportional to the foreign ownership share.

The high zero-holding probabilities reported in Table II (Panel C) warrant emphasis, because they are central to the large short-sale effects documented below. The short-sale constraint binds frequently in the estimated economy: Table II (Panel C) reports zero-holding probabilities of

77.7% and 73.7% for domestic institutions, 72.8% and 73.7% for domestic individuals, and 92.8% for foreign institutions in type-I; equivalently, the positive-holding probabilities are 22.3%, 26.3%, 27.2%, 26.3%, and 7.2%, respectively. The combination that looks most extreme—foreign institutions allocate their entire processing capacity to type-I data ($M_{K,(\mathrm{I})} = 2$) and yet hold type-I shares only about 7% of the time—is economically coherent rather than contradictory. Information has option value: it raises the precision of the posterior in the states in which the constraint is slack and the investor actually participates, and a low participation probability does not imply a low marginal value of data, because the marginal value is computed at the participation states. More generally, the information choice is evaluated ex ante over all states, whereas portfolio participation is determined ex post after observing signals and prices; the two decisions therefore need not move together. The high binding probabilities are a property of the estimated primitives (large position-size exposures relative to the aggregate supply noise), not an input. A referee could still reasonably ask whether the calibration makes the constraint unrealistically dominant; we discuss this calibration sensitivity in the Internet Appendix.

Fifth, cross-asset learning transmits the effect to both share types. Under partial access, foreign institutions can trade type-I but not type-II shares, yet their information problem is two-dimensional because the two fundamentals are correlated: type-II information affects the posterior relevant for type-I trading through the Schur-complement channel, and changes in foreign type-I trading alter the price-information environment faced by domestic investors, who trade and learn about both assets. The model therefore provides a natural explanation for why the effects of foreign access need not be confined to the shares that foreigners are permitted to hold.

Finally, foreign entry also changes the composition of the information-bearing investor population in the opposite direction. Foreign institutions have superior hard-information processing but no domestic soft information, so their entry simultaneously adds precise processed hard information and dilutes the average density of domestic soft information. The net effect on market outcomes is therefore not determined by the foreign ownership share alone; it depends on the interaction of the foreign mass, processing technology, soft-information precision, data allocation, asset access, and portfolio constraints.

These mechanisms operate at the margin, and their quantitative magnitude in the China calibration is moderate. In the constrained closed-to-partial comparison, price-based variance reduction changes from 8.85% to 9.32% for type-I (about a 5.3% increase relative to the closed-market level) and from 4.18% to 4.15% for type-II (Table III). The claim supported by the calibration is

therefore not that the benchmark produces large movements in every statistic, nor that foreign effects are disproportionately large relative to the 6.02% ownership share; it is that the effect is disproportionate relative to the 0.14% actively participating foreign mass: foreign institutions are informationally different—more efficient processors of hard information—and even a small active foreign mass can affect the public price signal and hence the decisions of the much larger domestic sector. Quantitatively, the foreign-entry effect on type-I price informativeness is positive but modest ($+0.47$ percentage points of price-based variance reduction, about 5.3% relative to the closed-market level). The much larger gains documented in the unrestricted counterfactual below arise primarily from relaxing the severe short-sale constraint, not from foreign ownership alone.

D. Counterfactual Market Opening

The benchmark explains how a relatively small foreign ownership share can affect the information environment under partial access. We next use the estimated model to ask what would happen if foreign institutions could trade both share types. Holding the estimated Chinese primitives fixed, we compare two scenarios: full access with a short-sale constraint maintained, in which foreign institutions can trade both Type-I and Type-II shares but all risky-asset positions remain nonnegative; and full access without a short-sale constraint. Table III reports the outcomes, together with the partial-access benchmark for comparison.

The comparison shows that full opening has sharply different implications depending on the portfolio regime. When short sale is allowed, the full-access equilibrium is well behaved: foreign institutions allocate all of their processing capacity to Type-I information, and prices remove 65.92% and 37.99% of prior variance about Type-I and Type-II fundamentals, respectively (Table III, Panel D). With a short-sale constraint, the constrained full-access equilibrium converges to a pure symmetric data-choice fixed point at the current calibration (Internet Appendix Section IA.H); foreign processing remains concentrated on Type-I data ($M_{K,(\mathrm{I})} = 2.000$ and $M_{K,(\mathrm{II})} = 0.000$), but the constraint severely impedes the aggregation of this private information into prices: price-based variance reduction falls to 9.32% for Type I and 4.15% for Type II, and total variance reduction is 83.97% and 66.81%.

The short-sale constraint therefore does more than restrict foreign portfolios; it changes the private value of information. The calibration identifies the short-sale margin as the dominant

Table III
Market-access and short-sale counterfactuals.

All counterfactuals hold estimated Chinese primitives fixed. Closed removes foreign investors; constrained full access permits foreign Type-II trading but retains a short-sale constraint; Panel D also permits short sale. Panel E reports constrained-minus-unrestricted differences in percentage points. Both full-access equilibria converge; entries below 0.0005 in absolute value are reported as zero.

	Closed Counterfactual	**Partial (Estimated)**	Full access† (Constrained)
Panel A. Information production			
Domestic $M_{I,(\mathrm{I})}$	2.000	2.000	2.000
Domestic $M_{I,(\mathrm{II})}$	0.000	0.000	0.000
Foreign $M_{K,(\mathrm{I})}$	n.a.	2.000	2.000
Foreign $M_{K,(\mathrm{II})}$	n.a.	0.000	0.000
Panel B. Price-based variance reduction			
Type I, VR^{p}_{I}	8.85%	9.32%	9.32%
Type II, VR^{p}_{II}	4.18%	4.15%	4.15%
Panel C. Total variance reduction			
Type I, $VR^{\mathrm{tot}}_{\mathrm{I}}$	82.32%	83.97%	83.97%
Type II, $VR^{\mathrm{tot}}_{\mathrm{II}}$	67.23%	67.23%	66.81%
Panel D. Full access, short sale allowed			
Foreign capacity allocated to Type I			100.0%
Foreign capacity allocated to Type II			0.0%
Type I, VR^{p}_{I}			65.92%
Type II, VR^{p}_{II}			37.99%
Type I, $VR^{\mathrm{tot}}_{\mathrm{I}}$			87.60%
Type II, $VR^{\mathrm{tot}}_{\mathrm{II}}$			72.04%
Panel E. Effect of a short-sale constraint (constrained minus unrestricted, pp)			
$\Delta VR^{p}_{\mathrm{I}}$	−51.56	−56.60	−56.60
$\Delta VR^{p}_{\mathrm{II}}$	−34.71	−33.84	−33.84
$\Delta VR^{\mathrm{tot}}_{\mathrm{I}}$	−3.56	−3.63	−3.63
$\Delta VR^{\mathrm{tot}}_{\mathrm{II}}$	−5.29	−5.11	−5.23

†Full access with a short-sale constraint maintained.

driver of the large full-opening numbers: expanding investability from partial to full access leaves price informativeness essentially unchanged within either trading regime (constrained full access delivers the same $9.32\%/4.15\%$ price-based variance reductions as the partial-access benchmark, and the unrestricted partial- and full-access equilibria are likewise indistinguishable), whereas removing the short-sale constraint raises Type-I price-based variance reduction from 9.32% to 65.92%. The calibration also draws a sharp distinction between information possessed by investors and information revealed through equilibrium prices: a market can contain substantial private information while prices reveal only a small fraction of it when portfolio constraints weaken informed risk bearing. Portfolio constraints primarily impede the aggregation of private information into prices rather than altering the concentrated Type-I information allocation itself. Market access and trading flexibility jointly determine information aggregation, but the two margins are not symmetric complements: for type-I the closed-to-partial foreign-entry effect on price-based variance reduction is $+0.47$ percentage points under the short-sale constraint and approximately $+5.51$ percentage points without it (an interaction of about $+5.04$ percentage points), whereas for type-II the analogous interaction is approximately -0.87 percentage points and incremental partial-to-full investability itself is essentially zero in either trading regime. The quantitative weight therefore falls on the portfolio-constraint margin at the Chinese benchmark. We next evaluate the efficiency of the resulting decentralized information allocation.

E. Information Production Efficiency

For compactness, the tables report variance reduction relative to the prior. Price-based variance reduction uses the posterior based on the price signal alone, while total variance reduction uses a population-wide posterior-precision statistic with a common population denominator. This quantitative statistic is distinct from the asset-eligible average posterior precision $\bar{\hat{\Omega}}_{y,(N)}$ used in the analytical comparative statics above. For the population-wide statistic, full symmetric individual posterior precision matrices are averaged before inversion; both variance-reduction statistics invert a full precision matrix before taking its diagonal because dividend innovations are correlated.

Table IV
Decentralized and efficient information production.

The planner chooses a common institutional allocation of capacity $\bar{M}$; prices and private portfolios are re-solved under each row's trading rules. Welfare is the investor-mass-weighted sum of ex ante certainty equivalents. Solutions are checked on a global grid, allocation boundaries, and local refinement. Full $+c$ is the converged constrained full-access equilibrium.

Panel A. Information-production wedges

Regime	$\Delta M_{I,\mathrm{I}}$	$\Delta M_{I,\mathrm{II}}$	$\Delta M_{K,\mathrm{I}}$	$\Delta M_{K,\mathrm{II}}$
Closed	-2.000	$+2.000$	n.a.	n.a.
Closed $+c$	-2.000	$+2.000$	n.a.	n.a.
Partial	-2.000	$+2.000$	-1.944	$+1.944$
Partial $+c$	-2.000	$+2.000$	-1.872	$+1.872$
Full	-2.000	$+2.000$	-1.950	$+1.950$
Full $+c$	-2.000	$+2.000$	-1.872	$+1.872$

Panel B. Welfare

Regime	Decentralized W_r	Planner $W_{r,\mathrm{pl}}$	Wedge ΔW_r
Closed	0.8382	0.8898	0.0516
Closed $+c$	0.6502	0.6562	0.0060
Partial	0.8011	0.8923	0.0912
Partial $+c$	0.6440	0.6514	0.0074
Full	0.8018	0.8931	0.0913
Full $+c$	0.6490	0.6561	0.0071

F. *Which friction drives the quantitative result?*

The counterfactual comparison shows that, at the China calibration, relaxing the short-sale constraint has a much larger effect on price informativeness than expanding foreign investability from partial to full access. The Internet Appendix reports friction-removal exercises that separately shut down the soft-information advantage and the processing-technology gap. Those exercises show that the quantitative result reflects the interaction of asset access with the technology–soft-information endowment, rather than investability in isolation.

G. *Reduced-Form Evidence*

The quantitative analysis above characterizes how foreign access affects equilibrium information allocation and prices. This subsection examines whether the Shanghai–Hong Kong Stock Connect announcement/launch is accompanied by differential movements in market outcomes for type-I and type-II stocks. The purpose is not to identify the structural mechanism, but to provide descriptive empirical evidence complementary to the structural analysis.

G.1. Empirical Design

Because market participants could anticipate the policy change before trading started, the design distinguishes the announcement-to-launch interval from the post-launch period and estimates

$$Y_{it} = \alpha_i + \gamma_{j(i),t} + \beta_A(\text{Eligible}_i \times \text{Announcement}_t) + \beta_L(\text{Eligible}_i \times \text{Post}_t) + \varepsilon_{it}, \tag{18}$$

where α_i denotes a firm fixed effect and $\gamma_{j(i),t}$ an industry-by-day fixed effect. The eligible group is fixed using the April 10, 2014 reference universe and retained throughout the sample. The preferred comparison restricts both groups to Shanghai-listed stocks: 562 eligible stocks and 443 never-eligible stocks. This same-exchange comparison avoids confounding the treatment contrast with Shanghai–Shenzhen movements. The preferred specification omits contemporaneous firm controls, which can respond to liberalization and are mechanically related to return-based outcomes.

The outcome is idiosyncratic volatility, the annualized 30-trading-day standard deviation of market-model residuals. β_A is the announcement-to-launch differential relative to the pre-announcement baseline, β_L is the post-launch differential, and $\beta_L - \beta_A$ is the incremental post-

launch contrast. These return-based measures are not direct empirical counterparts of the model's precision objects; the exercise documents differential market outcomes around the opening rather than tests the structural mechanism.

G.2. Regression Evidence and Interpretation

Table V reports the preferred same-exchange specification and a size-adjusted variant. In the preferred specification, eligible stocks have lower idiosyncratic volatility both between announcement and launch (-0.0388, $s.e. = 0.0066$) and after launch (-0.0385, $s.e. = 0.0065$), while the incremental post-launch contrast is indistinguishable from zero. Size adjustment attenuates the announcement-period estimate but leaves the post-launch differential negative. Thus the broad differential is robust, whereas its timing is not cleanly attributable to anticipation versus implementation.

Table V
Stock Connect and idiosyncratic volatility.

Estimates compare announcement-eligible with never-eligible Shanghai stocks using equation (18). Column (1) includes firm and industry-by-day fixed effects; column (2) additionally includes pre-treatment size-decile-by-day effects on the common 2013-size sample. Standard errors are two-way clustered by firm and date. The final row is $\beta_L - \beta_A$.

	(1)	(2)
Eligible × Announcement-to-launch period	-0.0388^{***}	-0.0148^{*}
	(0.0066)	(0.0080)
Eligible × Post-launch period	-0.0385^{***}	-0.0277^{***}
	(0.0065)	(0.0078)
Post-launch − Announcement	0.0002	-0.0129^{*}
	(0.0061)	(0.0073)
Sample	Shanghai	Shanghai
Firm fixed effects	Yes	Yes
Industry × day fixed effects	Yes	Yes
Pre-treatment size decile × day fixed effects	No	Yes
Controls	No	No
Clustering	Firm & date	Firm & date
Observations	761,168	723,108
Within R^2	0.0044	0.0015

The Internet Appendix reports event studies, alternative outcomes, placebo and window

checks, and a fully time-varying eligibility design. Those exercises support the broad-period differential but show that short-run timing and some alternative return outcomes are specification-sensitive. Accordingly, the reduced-form evidence is descriptive: it does not separately identify the model's information-production channels or provide a directional test of its price-precision predictions. The calibration instead quantifies those mechanisms under the maintained model.

VII. Conclusion

This final section synthesizes the theoretical and quantitative implications for market-access policy.

When foreign institutional investors enter the domestic market, they bring in better data-processing technology for hard information but face the lack of soft information. This mismatch can create information inefficiency, and the two opposite forces also reconcile the mixed empirical evidence on whether domestic investors have an advantage over foreign traders in domestic stock markets when domestic stock markets start to liberalize.

With partial market access, foreign institutional investors shift marginal data-processing capacity toward internationally non-tradable shares whenever the information spillover rate exceeds 1; the endpoint values of the spillover rate determine whether the optimum is a corner or an interior allocation. This cross-asset learning can generate information inefficiency on the market. The hard information obtained from processing data for internationally non-tradable shares is not aggregated into their equilibrium price but partially aggregated into the equilibrium price of foreign investable shares given that the two types of shares are correlated. Thus a distortion of price informativeness is created. This distortion is further strengthened when hard information of eligible companies comes with higher precision.

More generally, market access, investors' private information, and price informativeness need not move together. Making an asset newly investable can reduce its price informativeness when it eliminates cross-price information that previously helped reveal its fundamentals; conversely, under partial access, an increase in price informativeness can coincide with a decline in eligible investors' average posterior precision.

In the case of China, foreign institutional investors allocate all of their data-processing capacity to foreign-investable shares, and domestic institutions do the same. Price-based variance reduction for investable shares rises from 8.85% to 9.32% when the market opens to foreign in-

vestors, while that for non-investable shares is essentially unchanged (4.18% to 4.15%). In the full market access counterfactual, the effects depend sharply on portfolio constraints: without a short-sale constraint, price-based variance reduction rises to 65.92% for investable and 37.99% for non-investable shares, whereas maintaining the short-sale constraint leaves the price-based reductions close to the partial-access values (9.32% and 4.15%).

These quantitative results are conditional implications of the calibrated model rather than globally identified causal facts. The benchmark is a locally selected equilibrium in which the short-sale constraint binds frequently and foreign institutions have a substantial processing advantage; the type-II effective-precision parameter is set-identified at the allocation corner, and an economically distinct equal-precision specification fits almost as well (objective 1.2919 versus 1.2587) while flipping the domestic information allocation (Internet Appendix). Within this locally selected benchmark, however, the central margin is robust: relaxing the short-sale constraint, not incremental investability, drives the large gains in price-information aggregation.

Appendix: Proofs of Main Theoretical Results

This appendix supplies proofs of the paper's three main theoretical results. It covers the information-allocation proposition under heterogeneous market access (Proposition 1), the intermediate-precision cross-asset-learning proposition (Proposition 2), and the price-informativeness proposition (Proposition 3). Supporting lemmas, the constrained-equilibrium characterization, welfare derivations, and quantitative details appear in the Internet Appendix.

A.1. Proof of Proposition 1

Proof of Proposition 1. For domestic institutions, Lemma 1 reduces the data-choice problem to maximizing posterior precision subject to the binding capacity constraint. The interior first-order condition equalizes posterior precision in effective-data units, and the endpoint comparisons give (9).

For foreign institutions under partial access, Lemma 2 reduces the objective to Type-I Schur-complement precision. With $M \equiv M_{k,(\mathrm{II})}$ and $M_{k,(\mathrm{I})} = \bar{M} - M$, the marginal objective is $\frac{d\hat{\Omega}^{S}_{kyt,(\mathrm{I})}}{dM} = \Pi_{e,(\mathrm{I})}[\kappa(M) - 1]$. Holding the equilibrium price signal fixed, $\kappa(M)$ is continuous and, whenever the cross-precision is nonzero, strictly decreasing. The two endpoint conditions therefore give the Type-I and Type-II corners, while $\kappa(0) > 1 > \kappa(\bar{M})$ yields the unique interior solution $\kappa(M^{\text{int}}) = 1$. These are exactly the cases in (10). □

A.2. Proof of Proposition 2

Proof of Proposition 2. At the candidate all-Type-I foreign-processing corner, the foreign allocation rule is governed by the marginal spillover rate $\kappa_E(0)$. As $E \to 0$, the direct precision contribution of a Type-II observation vanishes, so $\kappa_E(0) \to 0$ and the Type-II allocation is zero. At the other extreme, equilibrium price precision rises sufficiently quickly with E that the cross-precision ratio z_E vanishes faster than the direct precision gain increases: $z_E^2 E \to 0$. Thus $\kappa_E(0) \to 0$ also as $E \to \infty$, and the Type-II allocation is again zero.

Internet Appendix Section IA.A.4.2 constructs a primitive lower bound for $\kappa_E(0)$ that is single-peaked in E. Under the stated sufficient condition, that lower bound exceeds one on a nonempty bounded interval. Hence $\kappa_E(0) > 1$ throughout that interval, the all-Type-I corner violates the foreign allocation rule, and strictly positive Type-II processing is optimal. The full

bounds, high-precision case split, and lower-bound derivation are provided in that Internet Appendix section. □

A.3. Proof of Proposition 3

Proof. **Part (i).** The Type-II price-precision decomposition in Internet Appendix Section IA.D separates the change in the own-price component from the cross-price component available under partial access. Thus Type-II price informativeness falls if and only if

$$\left(\zeta^2_{(\mathrm{II});\mathrm{fu}} - \zeta^2_{(\mathrm{II})}\right) \Omega_{x,(\mathrm{II})} < \zeta^2_{(\mathrm{I}\leftarrow\mathrm{II})} \Omega_{x,(\mathrm{I})}.$$

Part (ii). At the all-Type-I foreign-processing corner, the Type-I SNR change has the same sign as the net change $\mathcal{H}_{\mathrm{I}}$ in aggregate Type-I hard-information precision. The posterior-precision accounting derived in the Internet Appendix is

$$\Delta\bar{\hat{\Omega}}_{y,(\mathrm{I})} = A_{\mathrm{I}}\mathcal{H}_{\mathrm{I}} - \frac{\lambda_K}{\lambda_2} C_{\mathrm{I}} - \mathcal{Q}_{\mathrm{I}},$$

where $A_{\mathrm{I}}, C_{\mathrm{I}} > 0$ and $\mathcal{Q}_{\mathrm{I}} \geq 0$ is the foreign portfolio-constraint discount. If Type-I price informativeness weakly falls, then $\mathcal{H}_{\mathrm{I}} \leq 0$, making the first and third terms nonpositive and the dilution term strictly negative. Average Type-I posterior precision therefore falls strictly. Conversely, when price informativeness rises, the joint outcome in the proposition occurs whenever

$$0 < \mathcal{H}_{\mathrm{I}} < \frac{(\lambda_K/\lambda_2) C_{\mathrm{I}} + \mathcal{Q}_{\mathrm{I}}}{A_{\mathrm{I}}}.$$

The left inequality makes Type-I price informativeness rise, while the right inequality makes average Type-I posterior precision fall. The full rearrangement is in the Internet Appendix. □

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~~Online Appendix~~Internet Appendix

The complete derivations, proofs, and institutional background of this paper ~~are collected in the Online Appendix, submitted as a separate file (`market_opening_JF_online_appendix.tex`). Appendix sections are lettered A–G in both documents~~are collected in the Internet Appendix, submitted as a separate file following the paper, with sections numbered IA.A, IA.B, ...; all cross-references from the main text, including the Internet Appendix Sections IA.F and IA.H and the price-coefficient formulas IA.26, IA.27, IA.28, and IA.29, resolve to the ~~Online Appendix~~Internet Appendix.